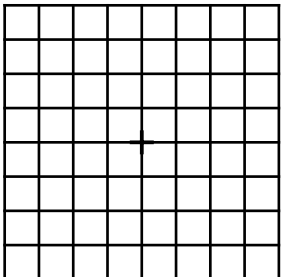
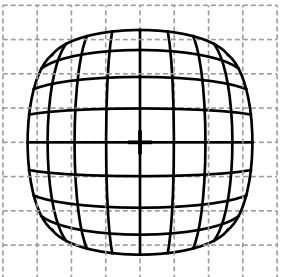
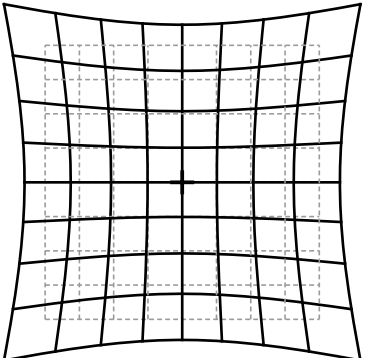
(a) Undistorted

(b) Barrel, $k_1 < 0$

(c) Pincushion, $k_1 > 0$